

Dibyendu “Dev” Nath

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Engineering Leader with 11+ years scaling mission-critical AI and Ads infrastructure at Google.

- **AI Serving:** Leads enterprise Gemini API serving infrastructure on Google Cloud.
- **Ads Infrastructure:** Previously led low-latency Shopping Ads serving on google.com.
- **Leadership:** Specializes in accelerating 0-to-1 product launches and scaling high-performing teams.

EXPERIENCE

Google LLC – San Francisco Bay Area, CA
Senior Engineering Manager / Tech Lead

Aug 2015 – Present

Gemini Enterprise Agent Platform – Google Cloud

Mar 2025 – Present

- Senior Engineering Manager / Tech Lead for Gemini Enterprise Agent Platform (formerly Vertex AI).
 - Delivering low-latency, multi-region serving of Gemini models for Google Cloud enterprise customers.
 - Accountable for service reliability and latency; drove 10x reduction in downtime for premium customers by achieving a 99.99% availability SLO (up from 99.9%).
 - Leading engineering for new Gemini model launches on the platform, shipping same-day as the rest of Google, with enterprise compliance hardening and external SLAs.
 - Building external APIs (GenerateContent and the new Interactions API), along with safety controls, quota, isolation and scaling policy for these models, sustaining availability SLOs through 3x token growth in three quarters.
 - Shaped the platform’s model and agent API strategy, taking the Interactions API from proposal to launch.
 - Partnering with Google DeepMind research teams to take new models and agents (e.g. CodeMender) from research to enterprise production.
- Rearchitected the multimodal serving path (image, video, audio, PDF) for high throughput, adding efficient load balancing and sandboxed media processing that keeps latency low while isolating untrusted content.
- Grew team from 8 to 25+ in less than a year. Currently leading a 30+ person org spanning Sunnyvale and Seattle, with 3 teams led by engineering managers, focusing on execution rigor and talent growth.
- Instituted review and launch practices that improved model launch velocity from ~1 month to ~1 week; drove operational excellence, incident readiness and postmortem culture.

Google Ads Shopping Serving

Aug 2015 – Mar 2025

- Led the Google Ads Shopping Serving engineering team.
 - Owned low-latency backend serving and indexing infrastructure for Product Listing Ads on google.com, powering \$XXB+ of annual ad revenue.
- Drove a serving-stack redesign to absorb 40% annual growth in shopping offer inventory (XX billions of offers).
 - Cut end-to-end data staleness by 54% on average and 91% for high-impression offers, the redesign’s key goal, while serving XXX kQPS at under 250ms latency.
 - Recognized with a company-wide Google Tech Impact award in 2023.
- Joined as a new-grad engineer and advanced to Tech Lead / Manager; built a high-performing team through hiring, mentoring, and growth planning, alongside deep technical contributions.

University of California, Santa Barbara – Santa Barbara, CA

Sep 2013 – Sep 2015

Research Assistant, RACE Lab – StochSS : Cloud-based Stochastic Simulation as a Service

- Built and maintained StochSS, a cloud service for running stochastic simulations across heterogeneous compute resources; published in PLOS Computational Biology.

AppFolio Inc. – Goleta, CA

Jun 2014 – Sep 2014

Software Engineering Intern – RentMatch : Appfolio’s Pricing Analytics (Data Science) team

- Built machine learning models to score rental unit similarity using text, amenities and census-derived features.
- Developed a distributed data collection and processing framework for running complex data joins across data centers.

McAfee Inc. – Bangalore, India

Feb 2012 – Aug 2013

Software Development Engineer – Endpoint Encryption for Files and Folders (EEFF) team

- Delivered features across client endpoints and policy management server for enterprise endpoint encryption product.
- Built ‘Kill Pill’ prototype for remote deactivation and secure wiping of encrypted removable USB devices.

EDUCATION

University of California, Santa Barbara – Santa Barbara, CA

Sep 2013 - Jun 2015

Master of Science, Computer Science

Advisors: Prof. Chandra Krintz, Prof. Rich Wolski

West Bengal University of Technology – Kolkata, India

Aug 2007 - Jul 2011

Bachelor of Technology, Computer Science & Engineering

SKILLS

- Team and org leadership: vision & strategy planning, building roadmaps, day-to-day project management, execution rigor and incident management with a focus on org health.
- Building and growing teams; coaching engineers to senior/staff; fostering a strong engineering culture.
- Distributed and parallel computing, high-throughput serving, low-latency optimization, streaming data pipelines, LLM serving/inference platforms, large-scale indexing and caching systems.
- *Programming*: Extensive experience in C++, C, Python. Proficient in Java, SQL, Go, shell scripts, JavaScript.
- *Protocols & Frameworks*: Protocol Buffers, Stubby/gRPC, HTTP, MapReduce/Flume/Hadoop, memcached/redis.

PUBLICATIONS

Shanbhogue, M., Li, Z., Zhang, S., Hernández Ábrego, G., Huang, S.C., Jain, A., et al. (incl. Nath, D.) 2026. **“Gemini Embedding 2: A Native Multimodal Embedding Model from Gemini.”** arXiv preprint arXiv:2605.27295.

Drawert, B., Hellander, A., Bales, B., Banerjee, D., Bellesia, G., Daigle Jr, B.J., Douglas, G., Gu, M., Gupta, A., Hellander, S., Horuk, C., Nath, D., et al 2016. **“Stochastic Simulation Service: Bridging the Gap between the Computational Expert and the Biologist.”** PLOS Computational Biology, 12(12), p.e1005220.

D. Nath, S. Ray, S.K. Ghosh, **“Fingerprint Recognition System: Design & Implementation,”** Proceedings of International Conference on Scientific Paradigm Shift In Information Technology & Management, SPSITM’11, January, 2011.

AWARDS & HONORS

- Recipient of company-wide **Google Tech Impact Award** in 2023 and **Google Commerce Tech Impact Award** in 2022 for technical contributions & leadership.
- Ranked **585th (99.57 percentile)** out of about 130,000 students in *Computer Science*, in *Graduate Aptitude Test in Engineering, 2011* (Indian Graduate School Admission Exam for IITs/NITs etc).
- Recipient of **National Merit Scholarship** for securing **rank 49** in *State Secondary Examination, 2005* among about 700,000 students.